Ian M. Nesbitt

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National Center for Ecological Analysis and Synthesis, U.C. Santa Barbara
DataONE Federation Software and Network Operations

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Education

2017-2022 **Masters Student in Earth and Climate Sciences**, *School of Earth and Climate Sciences, College of Natural Sciences, Forestry, and Agriculture, University of Maine*, Orono, ME, USA
GPA: 3.97
Thesis: Using diverse methods to determine the volume and stratigraphy of sediment in a formerly glaciated lake in central Maine, USA
Advisor: Seth W. Campbell, *Co-advisor*: Sean M.C. Smith
Associated Projects: readgssi, a Python tool to read and process radar data; SeidarT, a seismic and radar survey modeling toolbox

2009–2013 **Bachelor of Arts in Geosciences with Honors**, *Williams College*, Williamstown, MA, USA
Thesis: A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range
Advisor: David P. Dethier

2006–2009 **Degree with Honors**, *Holderness School*, Holderness, NH, USA

Awards and Scholarships

2021 Golden Key (invited)
2020 US Antarctica Service Medal
2013 Sigma Xi induction
2012 Williams College Class of 1960 Scholar in Geosciences

Volunteer Experience

Since 2025 Co-Chair, Earth Science Information Partners (ESIP) schema.org Cluster with Colin Smith
Since 2022 Nordic Ski Coach, Mt. Greylock Regional School
2018–2023 Co-President, Williams College Class of 2013
2017–2018 Co-Director of Reunion Planning, Williams College Class of 2013

Work Experience

Since 2022 **Research Software Engineer**, *National Center for Ecological Analysis and Synthesis (NCEAS), University of California Santa Barbara*, Santa Barbara, CA, USA *(remote)*
Duties: Python and Java development of open-source computing tools to interpret, quality-check, archive, and serve large quantities of research science data. Working on DataONE Federation metadata interoperability and data processing tools. DataONE Network Operations duties include monitoring and troubleshooting of DataONE member nodes and coordinating with the community to ensure the health of the network.
Supervisor: Matt Jones

Since 2022 **Assistant Head Coach**, *Berkshire Nordic Ski Club*, Williamstown, MA, USA
Duties: Group and one-on-one summer Nordic ski training, technique instruction, goal setting, customized training plans. Planning and coordination with other coaches, athletes, and parents.
Head Coach: Hilary Greene

2022 **Geospatial Engineering Consultant**, *Muddy Water Initiative*, Boston, MA, USA
Duties: Geospatial database management, design of SQL joins and relates, QA/QC and digitization of field data, presentation-quality map design, client and stakeholder interaction for Caroline Reeves.

2021 **Geophysicist**, *International Thwaites Glacier Collaboration, Project GHC*, Orono, ME, USA
Duties: Processed ground-penetrating radar (GPR) survey data from the Hudson Mountains, West Antarctica, in collaboration with John Woodward, Joanne S. Johnson, Brent Goehring, and Seth W. Campbell.

2021 **Research Software Developer**, *University of Maine*, Orono, ME, USA
Duties: Tested and debugged code for geophysical modeling software SeidarT, developed a multi-platform install script for Christopher C. Gerbi and Steven P. Bernsen.
Projects: https://github.com/UMainedynamics/seidart

2019–2020 **Lead Geophysicist**, *Allan Hills Antarctic Expedition I-165-M, Princeton University - NSF Award Number 1744993*, Allan Hills Field Camp, East Antarctica
Duties: Collected and processed 100 km of ground-penetrating radar (GPR) survey data from a blue ice area on the eastern side of the Trans-Antarctic Mountains. Deep field survival, data collection, processing, use of snow machines in harsh conditions.

2018–2020 **Lead Scientist**, *Raspberry Shake, S.A.*, Boquete, Chiriqui, Panama
Duties: Technical support, software development, QA/QC of seismic instruments, development and QA/QC of educational materials.
Projects: https://github.com/raspishake/rsudp, https://edu.raspberryshake.org

2014–2017 **Geophysical Scientist**, *e4sciences LLC*, Newtown, CT, USA
Duties: Led field team for many types of geophysical survey including Mobile LiDAR, ground-penetrating radar, high-precision GPS, multibeam echosounding, sub-bottom seismic, sidescan sonar. Field interpretation, and on-the-ground operations decision-making including boat handling.
Projects: Created survey-to-delivery workflow for 3D LiDAR end-user deliverables.

2013–2014 **Assistant Nordic Ski Coach**, *St. Michael's College*, Colchester, VT, USA
Duties: Created and supervised individualized training plans for athletes, technique instruction, race day logistics including ski preparation.

Teaching

2019 **Ground-Penetrating Radar Theory and Applications to Geological Mapping – SC2**, *Northeastern Section Meeting, Geological Society of America*, Portland, ME, USA
Seth W. Campbell and Ian M. Nesbitt (filling in for Steven A. Arcone) taught short course at NEGSA annual meeting, covering radar application, data collection, processing, and interpretation.

2018 **Global Environmental Change – ERS 201**, *University of Maine*, Orono, ME, USA
Karl J. Kreutz and Ian M. Nesbitt instructed students in accepted and debated climate theory from an earth systems perspective.

2017 **Earth Systems – ERS 200**, *University of Maine*, Orono, ME, USA
Aaron E. Putnam, Sean M.C. Smith, Peter O. Koons, and Ian M. Nesbitt instructed students in earth systems theory.

2012–2013 **GIS and Remote Sensing – GEOS 214**, *Williams College*, Williamstown, MA, USA
David P. Dethier and Ian M. Nesbitt instructed students in traditional and emerging geomorphologic theory, methods, and practice.

2012 **Geomorphology – GEOS 201**, *Williams College*, Williamstown, MA, USA
David P. Dethier and Ian M. Nesbitt instructed students in geographical information systems (GIS) theory, technology and software, and advised in the application of these methods via end-of-term projects.

Peer-Reviewed Publications

in review **The A-KBS Arctic Knowledge Based System: Science Gateway Integration for Exascale Arctic Data Processing and Geospatial Feature Prediction**, *ESS Open Archive*
Meisam Shayeghmoradi, Andrew Wilcox, Sheridan Parker, Aaron Bergstrom, Ian M. Nesbitt, Naima Kaabouch, and Timothy Pasch
preprint: 10.22541/essoar.177292734.41397214/v1

2026 **Broadly stable atmospheric CO2 and CH4 levels over the past 3 million years**, *Nature*
J. Marks-Peterson, Sarah Shackleton, John Higgins et al.
doi: 10.1038/s41586-025-10032-y

2026 **The arctic knowledge-based system: Science gateway integration for petascale arctic data processing and geospatial feature prediction**, *Applied Computing and Geosciences*
Andrew Wilcox, Meisam Shayeghmoradi, Stephen Miller, Ian M. Nesbitt, Saisri Pogalla, Aymane Ahajjam, Walker McKee, Sheridan Parker, Matthew Johnson, Aaron Bergstrom et al.
doi: 10.1016/j.acags.2026.100322

2025 **Assessing the suitability of sites near Pine Island Glacier for subglacial bedrock drilling aimed at detecting Holocene retreat–readvance**, *The Cryosphere*
Joanne S. Johnson, John Woodward, Ian M. Nesbitt, Kate Winter, Seth W. Campbell, Kier Nichols, Ryan Venturelli, Scott Braddock, Brent Goehring, Brenda L. Hall et al.
doi: 10.5194/tc-19-303-2025

2025 **Miocene and Pliocene ice and air from the Allan Hills blue ice area, East Antarctica**, *Proceedings of the National Academy of Sciences*
Sarah Shackleton, Valens Hishamunda, L. Davidge, Ed Brook, J. Marks-Peterson, Austin Carter, S. Aarons, Andrei Kurbatov, Douglas Introne, Y. Yan, Ian M. Nesbitt et al.
doi: 10.1073/pnas.2502681122

in review **The sediment delivery continuum from deglaciation to the modern watershed based on lake sedimentary deposits in the Northeastern USA**, *Quaternary Research*
Ian M. Nesbitt, Seth W. Campbell, Sean M.C. Smith, Bess G. Koffman, Steven A. Arcone, and Kristin M. Schild
preprint: 10.13140/RG.2.2.11567.05281

2021 **rsudp: A Python package for real-time seismic monitoring with Raspberry Shake instruments**, *Journal of Open Source Software*
Ian M. Nesbitt, Richard I. Boaz, and Justin Long
doi: 10.21105/joss.02565

2020 **Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures**, *Science*
Thomas Lecocq, Stephen P Hicks, Koen Van Noten, Kasper van Wijk, Paula Koelemeijer, Raphael SM De Plaen, Frédérick Massin, Gregor Hillers, Robert E Anthony, Maria-Theresia Apoloner, ... Ian M. Nesbitt et al.
doi: 10.1126/science.abd2438

2016 **Tracing subarctic Pacific water masses with benthic foraminiferal stable isotopes during the LGM and late Pleistocene**, *Deep-Sea Research Part II: Topical Studies in Oceanography*
Mea S. Cook, A. Christina Ravelo, Alan C. Mix, Ian M. Nesbitt, and Nari V. Miller
doi: 10.1016/j.dsr2.2016.02.006

Conference proceedings, presentations, and datasets

2025 **Validation and interoperability of schema.org as a science metadata standard: a DataONE perspective**, *Earth Science Information Partners (ESIP) Annual Meeting*, Seattle, WA, USA, Invited talk
Ian M. Nesbitt
Speaker page

2022 **Post-glacial sediment delivery continuum to an impounded valley reach in central Maine: a multi-disciplinary approach**, *Geological Society of America Abstracts with Programs*, Invited talk
Ian M. Nesbitt, Sean M.C. Smith, Seth W. Campbell, Bess G. Koffman, Steven A. Arcone, and Kristin M. Schild
doi: 10.1130/abs/2022NE-374798

2021 **Unprecedented present deglaciation of Pine Island Glacier Inferred from Ice Penetrating Radar Studies of Localised Ice Domes in the Hudson Mountains**, *International Thwaites Glacier Collaboration Conference*, Poster
John Woodward, Joanne S. Johnson, Seth W. Campbell, Ian M. Nesbitt, Brenda L. Hall, Scott Braddock, Meghan Spoth, Brent Goehring, Ryan Venturelli, Dylan H. Rood, Kier Nichols, Johnathan Adams, and Greg Balco

2019 **A decision-making framework for sedimentation analyses in dammed river corridor impoundments**, *Geological Society of America Abstracts with Programs 51*, Portland, ME, Talk
Ian M. Nesbitt, Sean M.C. Smith, Bess G. Koffman, Seth W. Campbell, and Steven A. Arcone
doi: 10.1130/abs/2019NE-328587

2018 **Sedimentary architecture and accumulation rates of multiple lakes in New England, USA**, *AGU Fall Meeting Abstracts 2018*, Washington, D.C., Poster NS41B-0830
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, Sean M.C. Smith, and Bess G. Koffman

2018 **Sedimentary architecture of Maine lakes derived with ground-penetrating radar**, *Maine Water and Sustainability Conference*, Augusta, ME, Poster
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith

2018 **Holocene sediment volume determined by ground-penetrating radar and sidescan sonar in Maine, USA**, *Geological Society of America, Northeastern Section - 53rd Annual Meeting*, Talk
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith
doi: 10.1130/abs/2018NE-311370

2017 **Using ground-penetrating radar and sidescan sonar to compare lake bottom geology in New England**, *AGU Fall Meeting Abstracts 2017*, New Orleans, LA, Talk PP44B-01
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith

2017 **New England lake bottom geology and sedimentation rates derived from ground-penetrating radar**, *GSA Annual Meeting*, Seattle, WA
Seth W. Campbell, Ian M. Nesbitt, Steven A. Arcone, Sean M.C. Smith
doi: 10.1130/abs/2017AM-306235

2015 **Geophysical imaging of dams and levees in the Northeast US**, *Society of Exploration Geophysicists Abstracts With Programs*, New Orleans, LA
Daniel A. Rosales, William F. Murphy IV, Matthew B. Art, Ian M. Nesbitt, Salvatore Triano, David C. Herron, Kurt Schollmeyer, James Trotta, W. Bruce Ward, Lisa Stewart, and William F. Murphy III

2014 **Tracing Bering Sea circulation with benthic foraminiferal stable isotopes during the Pleistocene**, *AGU Fall Meeting Abstracts 2014*, PP23D-08
Mea S. Cook, A. Christina Ravelo, Alan C. Mix, Ian M. Nesbitt, and Nari V. Miller

2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Thesis, Department of Geosciences, Williams College*, Williamstown, MA
Ian M. Nesbitt
Advisor: David P. Dethier
Available via Williams College Libraries
doi: 10.13140/RG.2.2.33197.20966

2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Geological Society of America, Northeastern Section - 48th Annual Meeting*, Bretton Woods, NH
Ian M. Nesbitt and David P. Dethier

2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Twenty-sixth annual Keck Research Symposium in Geology, proceedings 2013*, Pomona, CA
Ian M. Nesbitt and Robert Varga

Open-source software

2025 **MetacatUI 2.36.0**, *Arctic Data Center*
Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
https://github.com/NCEAS/MetacatUI
doi: 10.18739/a2hx15s97

2025 **MetacatUI 2.35.0**, *Arctic Data Center*
Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
https://github.com/NCEAS/MetacatUI
doi: 10.18739/a2q52ff87

2024 **HashStore — Python**, *Arctic Data Center*
DouMing Mok, Matthew Brooke, Jing Tao, Jeanette Clark, Matt Jones, and Ian M. Nesbitt
https://github.com/DataONEorg/hashstore
doi: 10.18739/a2zg6g87q

2024 **MetacatUI 2.34.0**, *Arctic Data Center*
Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
https://github.com/NCEAS/MetacatUI
doi: 10.18739/a2w669b2r

2021 **SeidarT: Seismic and radar survey modeling toolbox**
Steven P. Bernsen, Christopher C. Gerbi, Ian M. Nesbitt, Ann Hill, Senthil Vel, Knut Christianson, Seth W. Campbell, and Ben Hills
https://github.com/UMainedynamics/SeidarT
doi: 10.5281/zenodo.5498194

2020 **rsudp: Continuous visual display, sudden motion monitoring, and historical replay of Raspberry Shake data**
Ian M. Nesbitt, Richard I. Boaz, and Justin Long
https://github.com/raspishake/rsudp
doi: 10.21105/joss.02565

2019 **gpx2dzg: a tool to convert GPX files to GSSI's proprietary DZG format, and plot comparisons of marks in GPX and DZT/DZX files**, *Zenodo*
Developer and maintainer
https://github.com/iannesbitt/gpx2dzg
doi: 10.5281/zenodo.3260948

2019 **readgssi: an open-source tool to read and plot GSSI ground-penetrating radar data**, *Zenodo*
Developer and maintainer
https://github.com/iannesbitt/readgssi
doi: 10.5281/zenodo.1439119

Technical Skills

Languages	Python, R, Matlab, Node.js, Java, Ruby/Rails
Markup	Django Templates, HTML, CSS, LaTeX, Markdown, Overleaf
Platform Tools	Docker, Kubernetes, Selfhosted Linux servers, Raspberry Pi, Heroku, AWS
Developer Tools	GNU/Linux, Unix Terminal/bash/zsh, GNU Make, git, GitHub, Jupyter Notebooks, Visual Studio Code, Apple Developer Tools
Bug Tracking	Request Tracker, MantisBT, GitHub
APIs	DataONE API, FDSN, USGS Earthquakes/EarthExplorer, Telegram, Google Maps, Google Workspace
Other Tools	QGIS, ArcGIS, Inkscape, GIMP, Illustrator, Photoshop, ReflexW, RADAN
Communication	Slack, Discord, Mattermost
Office Suites	Microsoft Office, LibreOffice, Google Docs/Sheets/Slides

Spoken Languages

English	Native
Spanish	Beginner

Certifications

Since 2022	SafeSport Trained
2013, 2018	Wilderness First-Aid
2016	CPR, First-Aid, and Defibrillator
2019	Antarctic Deep Field Survival
2020	T3 Alliance certified instructor